

Contents

1 Reader-facing sound-change pilot	1
1.1 Included sections	1
2 Velar palatalization before front vowels	1
2.1 Historical discussion	1
2.2 Palatalization of k before front vowels and j (OEVelarPalatalizationKFront) . .	2
2.3 Velar palatalization before front vowels (OEVelarPalatalization)	2
3 The Old English i-umlaut and West Saxon palatal diphthongization	4
3.1 Historical discussion	4
3.2 Fronting under i-umlaut (OEIUmlautFronting)	4
3.3 Raising under i-umlaut (OEIUmlautRaising)	5
3.4 Diphthongal outcomes under i-umlaut (OEIUmlautDiphthong)	5
3.5 The composite i-umlaut rule (OEIUmlaut)	6
3.6 West Saxon palatal diphthongization (OEWsPalatalDiphthongization)	6
4 Nasal dissimilation	7
4.1 Historical discussion	7
4.2 Nasal dissimilation in short-vowel environments (OENasalDissimilation)	7
5 Reader-facing pilot source note	9
5.1 Style-model files consulted	9
5.2 Pilot source files consulted	9
5.2.1 General sound-change sources	9
5.2.2 Reference texts checked directly	9
5.3 Citation verification note	9
5.4 Scope of the pilot	9

1 Reader-facing sound-change pilot

This assembled pilot PDF is for checking the first reader-facing rewrite layer of the sound-change half.

1.1 Included sections

1. Velar palatalization before front vowels
2. The Old English i-umlaut and West Saxon palatal diphthongization
3. Nasal dissimilation

2 Velar palatalization before front vowels

2.1 Historical discussion

Luick places the change inside a broad early palatalizing movement. Under the heading “Frühe Verschiebungen in palataler Richtung,” he treats English k and g before bright vowels together with the larger field of palatal effects (Luick 1914–1940, 157, §168). His emphasis falls on the environment first: velars before bright vowels and in the vicinity of the palatal glide belong to one early phonological corridor. The examples associated with that corridor, such as *ceaster*, *geaf*, *giefan*, and *giest*, already show that consonantal palatalization and later vowel effects stand close together historically, even when they must be distinguished analytically (Luick 1914–1940, 157–67, §§168–182).

Campbell narrows the picture by distinguishing plain velars from the especially palatal-prone *sk* cluster. His remark that “[*sk*] is more prone to palatalization and assibilation than [*k*]” is brief, but it makes clear that different members of the larger palatal field need not behave identically

(Campbell 1959, 278, §440). Elsewhere in the same part of the grammar he uses forms such as *cild*, *dæg*, *giefan*, and *giest*, which show how palatalized velars, palatal influence, and later umlautal outcomes meet in the same region of the lexicon without collapsing into one process (Campbell 1959, 69–72, 89, §§170, 190–191).

Hogg makes the conditioning sharper still. He states that the change takes place when the velar consonant is adjacent to and in the same syllable as a front vowel or the palatal consonant *j* (Hogg 1992, 103–4). This formulation is important because it moves the discussion from a broad list of palatal outcomes to a more precise phonological environment involving adjacency and syllable structure.

Ringe and Taylor make the chronological relation still clearer. When they write that “after initial velars and *sk had been palatalized” West-Saxon diphthongization follows, plain velar palatalization becomes an earlier consonantal stage presupposed by later vowel developments (Ringe and Taylor 2014, 215, §6.5.1). Their own examples of the plain-velar rule, such as *weccan*, *licgan*, *leccan*, *secg*, *ecg*, *wicg*, and *brycg*, illustrate the same point in lexical detail: front vowels and *j* create the palatal environment in which plain *k* and *g* cease to behave as plain velars (Ringe and Taylor 2014, 213–14, §6.4.1).

Taken together, these accounts show a gradual tightening of focus. Luick treats palatalization as a broad early movement. Campbell distinguishes more sharply between plain velars and the *sk* complex. Ringe and Taylor then place the plain velar change in an explicit sequence that leads forward to later West-Saxon diphthongization. The literature therefore supports two claims at once: the change belongs to a larger palatalizing environment, and it must be kept distinct from neighboring processes if the sequence of developments is to be described accurately.

2.2 Palatalization of *k* before front vowels and *j* (OEVelarPalatalizationKFront)

The first part of the implementation isolates the *k*-side environments of the change.

```
define OEVelarPalatalizationKFront [
  {*k} -> {*tʃ} || .#. _ EnglishStarFrontVowel,
  {*k} -> {*tʃ} || _ [{*i} | {*ī}],
  {*k} -> {*tʃ} || _ {*í},
  {*k} -> {*tʃ} || [{*i} | {*ī}] _ EnglishStarFrontVowel,
  {*k} -> {*tʃ} || {*í} _ EnglishStarFrontVowel,
  {*k} -> {*tʃ} || [{*i} | {*ī}] _ .#. ,
  {*k} -> {*tʃ} || {*í} _ .#.
] .o. [
  {*k} {*k} -> {*tʃ} {*tʃ} || _ {*j}
] .o. [
  {*k} -> {*tʃ} || _ {*j}
] ;
```

In prose, the rule turns plain *k* into a palatal outcome before front vowels and *j*, including the geminated environment before *j*.

Historically, this section corresponds to the core of the older discussion of palatalized velars. It captures the environments behind forms such as *weccan*, *licgan*, and *leccan*, where front vowels or *j* trigger the palatal outcome in the first place (Ringe and Taylor 2014, 213–14, §6.4.1). It is also the part of the process that prepares forms later assumed by velar palatalization before front vowels (OEVelarPalatalization) and, farther on, by fronting under *i*-umlaut (OEIUmlautFronting).

Within the present implementation, this helper rule is not ordered separately from the broader velar-palatalization rule below. Its chronology is therefore that of the larger rule it feeds.

2.3 Velar palatalization before front vowels (OEVelarPalatalization)

The broader rule adds the *g* environments and composes them with the *k* palatalization rule above.

```

define OEVelarPalatalization [
  OEVelarPalatalizationKFront
] .o. [
  { *g } -> { *ǵ } || _ EnglishStarFrontVowel,
  { *g } -> { *ǵ } || EnglishStarFrontVowel _ .#.,
  { *g } -> { *ǵ } || EnglishStarFrontVowel _ EnglishStarFrontVowel,
  { *g } -> { *ǵ } || EnglishStarFrontVowel _ [EnglishStarConsonant - { *j }],
  { *g } { *g } -> { *ǵ } { *ǵ } || _ { *j }
] .o. [
  { *g } -> { *ǵ } || _ { *j }
];

```

In prose, the rule palatalizes plain *k* and *g* in front-vocalic and *j*-adjacent environments. Writing it as a separate rule clarifies the relative order of plain-velar palatalization, *sk*-palatalization, and umlautal developments.

The rule belongs after the earlier syncope that prepares forms like *stretch* and before the later umlautal rules that would otherwise over-palatalize forms such as *cow* and *lung*. See fronting under *i*-umlaut (OEIUmlautFronting) and the composite *i*-umlaut rule (OEIUmlaut) below.

If the rule is moved too early, before the syncope that prepares the consonant cluster, it breaks the derivation of *stretch*. With PGmc **strákkijanq* in the wrong order, the model produces *strecċan*; the expected Old English form is *streċċan*.

If it is moved too late, after *i*-umlaut, it over-palatalizes forms such as *cow* and *lung*. PGmc **kūi* then yields *cȳ*; the expected form is *cȳ*. PGmc **lúnganjō* yields *lunġen*; the expected form is *lun-gen*.

Once the rule is in place, plain velars before front vowels and *j* no longer remain plain. They become the palatal outcomes presupposed by later developments, including the umlautal rules discussed in fronting under *i*-umlaut (OEIUmlautFronting). That matters for dictionary-like forms such as *cild* or *dæg* and for the broader relation between consonantal palatalization and later vowel-fronting processes (Luick 1914–1940, 157, §168; Campbell 1959, 278, §440; Ringe and Taylor 2014, 203–15, §§6.4.1, 6.5.1).

The evidence places the rule within a wider palatalizing environment, but it does not require every neighboring palatal process to be merged with it. *sk* belongs to a related but distinct development, and the later umlautal material poses a different historical problem. The relation to the earlier syncope rule is likewise specific and limited: the *stretch* evidence shows a real dependency without turning the feeder process into a coequal sound law of the same scope.

3 The Old English i-umlaut and West Saxon palatal diphthongization

3.1 Historical discussion

Luick gives the change its traditional scale:

Der wichtigste Fall von palataler Beeinflussung ... war die Veränderung der urengischen Vokale durch i oder j der Folgesilbe.

(Luick 1914–1940, 166–67, §182)

Campbell gives the most compact classical formulation in English when he writes that “the process known as i-umlaut or i-mutation operates on practically all the sounds which it could theoretically affect in OE” (Campbell 1959, 69, §190). He immediately defines the core conditioning environment as a following i or j, and he goes on to trace the consequences across much of the vowel system, including forms such as *giest*, *giefan*, *hierde*, and *ieldra* (Campbell 1959, 69–72, §§190–197). Hogg continues in the same vein: “we come now to a change which is almost as uncontroversial as it is important” (Hogg 1992, 112). His examples, such as *bryd*, *trymman*, *bedd*, *ciest*, and *wiersa*, likewise emphasize that the change is a broad redistribution of vowel quality across the Old English vowel system (Hogg 1992, 112–14). Taken together, these statements leave little doubt that i-umlaut is one of the central Old English vowel changes.

The narrower palatal-diphthongal material is described differently. Ringe and Taylor treat West-Saxon diphthongization after initial palatals as a distinct process (Ringe and Taylor 2014, 215, §6.5.1), and Fulk is even more explicit about its chronological delicacy when he calls it “diphthongization by initial palatal consonants (which precedes front umlaut but not breaking)” (Fulk 2018, 74, §4.13). Ringe and Taylor’s examples such as *gieldan*, *scield*, and *scieppan* show that this narrower process is triggered by already palatal consonants and leads to specifically West-Saxon diphthongal outputs (Ringe and Taylor 2014, 215–16, §6.5.1).

The sequence of discussion is fairly clear. Luick, Campbell, and Hogg all give i-umlaut primary importance. Ringe and Taylor and Fulk then help separate that major change from the narrower West-Saxon diphthongization that stands beside it. The literature therefore establishes a large, system-wide umlautal change and a narrower adjoining process affecting words after initial palatals. That distinction matters because the two processes act in different environments and produce different lexical consequences.

3.2 Fronting under i-umlaut (OEIUmlautFronting)

The first component of the implementation handles the broad fronting of vowels under the influence of following i or j.

```
define OEIUmlautFronting [  
  {*a} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ā} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*e} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*o} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ō} -> {*ē} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*u} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ū} -> {*ȳ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*á} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*é} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ó} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,  
  {*ú} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger  
];
```

In prose, the rule fronts and raises the relevant simple vowels when a following i or j provides the trigger.

Historically, this is the most central part of the umlautal development described by Luick, Campbell, Hogg, Ringe and Taylor, and Fulk. Within the present implementation it stands after velar palatalization before front vowels (OEVelarPalatalization) and before the narrower West-Saxon palatal-diphthongization rule discussed below.

The handbooks describe the same conditioning environment in different ways but with the same phonological consequence: a following high front vocoid triggers the fronting of earlier back vowels. That is why forms such as *byre*, *fylgan*, *gylden*, *wyrm*, and *giest* can all be treated inside the same formal rule even though they belong to different lexical classes (Ringe and Taylor 2014, 222, §6.6.1; Campbell 1959, 69–72, §§190–191).

As a component rule, it shares the chronology of the composite i-umlaut rule (OEIUmlaut).

3.3 Raising under i-umlaut (OEIUmlautRaising)

The second component handles the raising of umlauted *æ* to *e*.

```
define OEIUmlautRaising [
  {*æ} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

In plain language, this rule takes the fronted low vowel created by the earlier fronting rule and raises it further where the same umlaut trigger still holds.

Historically, this belongs inside the same broad i-umlaut development. It is part of the same chronological development and shares the evidence base of the composite i-umlaut rule (OEIUmlaut).

This narrower subrule matters because the sources do not describe umlaut as simple fronting alone. Campbell explicitly notes that the low front vowel changes again before *m* and *n* in most dialects (Campbell 1959, 69, §190), and Hogg likewise treats short front vowels as part of the same assimilatory system (Hogg 1992, 112).

3.4 Diphthongal outcomes under i-umlaut (OEIUmlautDiphthong)

The third component handles the diphthongal outcomes that also undergo i-umlaut.

```
define OEIUmlautDiphthong [
  {*ea} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*ēa} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*io} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*īo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*eo} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*ēo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*éa} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*éo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
  {*ío} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

In prose, the rule states that diphthongal inputs are subject to umlaut as well: the vowel change is not confined to simple vowels.

This matters historically because the handbooks describe i-umlaut as a system-wide assimilatory development. The rule therefore stands inside the same chronological bracket as fronting under i-umlaut (OEIUmlautFronting) and raising under i-umlaut (OEIUmlautRaising), even though its outputs are shaped differently.

The relevant examples are the recurring West-Saxon *ie* forms cited in the handbooks, including *giest*, *giefan*, and *hierde* in Campbell and *ciest* in Hogg (Campbell 1959, 69–72, 78–80, §§190–191, 248–251; Hogg 1992, 112–14). The present formalization keeps those diphthongal outcomes visible as a distinct part of the general umlautal development and does not leave them implicit under the broad description of fronting.

3.5 The composite i-umlaut rule (OEIUmlaut)

The implementation also defines a composite rule that composes the three preceding parts.

```
define OEIUmlaut OEIUmlautFronting
  .o. OEIUmlautRaising
  .o. OEIUmlautDiphthong;
```

In prose, this says that the implementation treats the umlaut as a sequence of fronting, raising, and diphthongal adjustments composed in order.

Chronologically, the composite rule must follow velar palatalization before front vowels (OEVelarPalatalization). If it is moved too early, forms such as *cow* and *lung* become over-palatalized. PGmc **kūi* yields *cȳ*; the expected form is *cȳ*. PGmc **lūnganjō* yields *lungen*; the expected form is *lungen*.

Those failures show that the broad umlautal rule needs an earlier terminus post quem in the palatalization sequence, even though it remains the main vowel change within the present chapter.

The composite rule is important because the literature presents the umlaut as a single historical development even while the implementation decomposes it into formal parts. The composite definition is the point at which the separate fronting, raising, and diphthongal effects are treated as one chronological event in the Old English sequence.

3.6 West Saxon palatal diphthongization (OEWsPalatalDiphthongization)

The narrower West-Saxon rule is treated separately from the broader umlautal complex.

```
define OEWsPalatalDiphthongization [
  { *æ } -> { *ea } || .#. [ { *tʃ } | { *dʒ } | { *ʃ } | { *j } ] _ [EnglishStarConsonant | EnglishPa
  { *ǣ } -> { *ēa } || .#. [ { *tʃ } | { *dʒ } | { *ʃ } | { *j } ] _ [EnglishStarConsonant | EnglishPa
  { *e } -> { *ie } || .#. [ { *tʃ } | { *dʒ } | { *ʃ } | { *j } ] _ [EnglishStarConsonant | EnglishPa
  { *ē } -> { *īe } || .#. [ { *tʃ } | { *dʒ } | { *ʃ } | { *j } ] _ [EnglishStarConsonant | EnglishPa
  { *é } -> { *īe } || .#. [ { *tʃ } | { *dʒ } | { *ʃ } | { *j } ] _ [EnglishStarConsonant | EnglishPa
  { *ē } -> { *īe } || .#. [ { *tʃ } | { *dʒ } | { *ʃ } | { *j } ] _ [EnglishStarConsonant | EnglishPa
];
```

In prose, this rule diphthongizes certain vowels after already palatal consonants in West Saxon. It therefore has a narrower dialectal and chronological scope than the broader umlaut rule.

The historical evidence for that narrower scope is concrete. Ringe and Taylor illustrate the rule with forms such as *gielðan*, *sciold*, and *scieppan*, where an already palatal consonant triggers the diphthongal outcome (Ringe and Taylor 2014, 215–16, §6.5.1). Hogg's *giefan* and *sceap* material belongs to the same phonological zone (Hogg 1992, 108–9), while Fulk distinguishes this palatal-consonant-triggered diphthongization from the broad front-mutation process (Fulk 2018, 74, §4.13).

Its place is later than the composite i-umlaut rule (OEIUmlaut). If this rule is moved too early, the later ordering is constrained by forms such as *gift* and *sheath*. PGmc **gēftiz* then yields *ġieft*; the expected form is *ġift*. PGmc **skáiθiz* yields *scǣþ*; the expected form is *scēaþ*.

No tested lexical item provides a comparably precise later terminus ante quem. The available evidence therefore establishes the rule's relation to the earlier umlautal process much more clearly than it fixes a later point by which it must already have applied.

The two rules should accordingly be kept distinct. The broad umlautal rule accounts for a system-wide assimilatory change; the West-Saxon rule accounts for a narrower palatal-consonant-conditioned diphthongization whose chronological and dialectal scope is more restricted.

4 Nasal dissimilation

4.1 Historical discussion

Luick preserves individual outcomes such as “enitre ‘einjährig (aus *anwintri)” without isolating a separate law around them (Luick 1914–1940, 166). Campbell likewise reaches forms such as *heofon* in a discussion of suffixal variation and does not set them off in any special section on nasal dissimilation (Campbell 1959, 155). Hogg mentions *heofon* in the course of his account of back mutation, again without isolating a separate law (Hogg 1992, 112).

Fulk supplies the clearest general formulation: “In the cluster *mn*, the first consonant tends to lose its nasality by dissimilation, though the results are hardly regular” (Fulk 2018, 121, §6.11). Ringe and Taylor stay close to the lexical evidence and note that *enetre* reflects “loss of the second **n* by dissimilation” (Ringe and Taylor 2014, 282).

The discussion therefore develops from scattered lexical observations to a more explicit but still cautious generalization. Luick preserves the kind of form the rule is meant to capture. Campbell and Hogg show that related outcomes enter the handbooks, but only incidentally, as part of larger accounts of other changes. Fulk makes the recurrent *mn* tendency explicit, while Ringe and Taylor provide an exact lexical case in *enetre*. What emerges is a limited but recurring dissimilatory pattern whose scope is far smaller than that of the major Old English vowel laws.

4.2 Nasal dissimilation in short-vowel environments (OENasalDissimilation)

The implementation formalizes the change as a narrow rule applying in short vowel environments before a following *n*.

```
define OENasalDissimilation [  
    {*m} -> {*f} || EnglishStarShortVowel _ EnglishStarShortVowel {*n} [EnglishStarShort  
];
```

In plain language, the rule turns medial *m* into *f* in a restricted short-vowel environment before a following syllable containing *n*.

Historically, the rule captures the limited type of dissimilation reflected in forms such as *heofon*, *fæstenn*, and *enetre*. It is much narrower than the major vowel changes and is best understood as a recurring but partly lexicalized pattern.

The relation between the sources and the formalization is correspondingly close but not exact. Fulk formulates the tendency at the level of *mn* clusters and illustrates it with *heofon* and *fæstenn* (Fulk 2018, 121, §6.11). Ringe and Taylor show the same kind of development in *enetre* (Ringe and Taylor 2014, 282). Campbell’s “*heofon* is for older *hefzen*” and Hogg’s **hefon* > *heofon* preserve outcomes that the present implementation wants to keep visible (Campbell 1959, 155; Hogg 1992, 112). The formal rule is therefore narrower than the total set of handbook remarks: it models one plausible recurrent environment and does not claim to exhaust every dissimilatory development involving nasals.

Chronologically, the available tests do not identify a sharper position within the Old English sequence. When the rule is moved earlier, no lexical breakpoint appears before the inherited West-Germanic material that precedes the tested Old English changes. When it is moved later, the tests likewise fail to identify a more precise later boundary within the remainder of the Old English sequence.

No comparable pair of lexical failures fixes a narrower slot here. The present evidence therefore gives neither a precise terminus post quem nor a precise terminus ante quem for the rule within the tested sequence.

Even so, the rule has real interpretative consequences. It provides a place in the implementation for outcomes of the *heofon*, *fæstenn*, and *enetre* type discussed in the literature (Fulk 2018, 121, §6.11; Ringe and Taylor 2014, 282; Campbell 1959, 155; Luick 1914–1940, 166; Hogg 1992, 112). Without an explicit rule, those outcomes would be left to diffuse analogy or to unexplained exception lists.

The evidence points to a narrow dissimilatory tendency, especially in mn-type clusters and a small group of lexical outcomes. There is no support for a regular change operating across a broad phonological field. The rule is secure enough to model, but the available tests leave its position within the Old English sequence underdetermined.

5 Reader-facing pilot source note

5.1 Style-model files consulted

- Germanic/docs/lexeme_reports/writing_skill/README.md
- Germanic/docs/lexeme_reports/writing_skill/book_entry_template.md
- Germanic/docs/lexeme_reports/model_entries/2183-shoulder-sculdram.model.md

5.2 Pilot source files consulted

5.2.1 General sound-change sources

- Germanic/docs/sound_changes/change_reports/full/052-velar-palatalization-hinge.md
- Germanic/docs/sound_changes/change_reports/full/055-056-umlaut-core.md
- Germanic/docs/sound_changes/change_reports/full/058-oe-nasal-dissimilation-residual-note.md
- Germanic/docs/sound_changes/literature_dossiers/052-velar-palatalization-hinge.dossier.md
- Germanic/docs/sound_changes/literature_dossiers/055-056-umlaut-core.dossier.md
- Germanic/docs/sound_changes/literature_dossiers/058-oe-nasal-dissimilation-residual.dossier.md
- Germanic/docs/sound_changes/order_tests/chronology_cards/SC052-oe-velar-palatalization.md
- Germanic/docs/sound_changes/order_tests/chronology_cards/SC055-oe-i-umlaut.md
- Germanic/docs/sound_changes/order_tests/chronology_cards/SC056-oe-ws-palatal-diphthongization.md
- Germanic/docs/sound_changes/order_tests/chronology_cards/SC058-oe-nasal-dissimilation.md
- Germanic/fsts/germanic.txt

5.2.2 Reference texts checked directly

- docs/references/campbell_old_english_grammar.txt
- docs/references/hogg_vol1.txt
- docs/references/ringe_taylor_linguistic_history_vol2.txt
- docs/references/luick_historische_grammatik.txt
- docs/references/fulk_comparative_grammar_early_germanic.vision.txt
- docs/references/fulk_comparative_grammar_early_germanic.pdf

5.3 Citation verification note

For the revised pilot chapters, page citations were checked against the local text witnesses and, where useful, against the repository PDF witness as well.

5.4 Scope of the pilot

The pilot is intentionally small. It tests reader-facing section design, quotation method, code presentation, and chronology explanation before any full-volume rewrite is attempted.

Campbell, Alistair. 1959. *Old English Grammar*. Clarendon Press.

Fulk, R. D. 2018. *A Comparative Grammar of the Early Germanic Languages*. Vol. 3. Studies in Germanic Linguistics. John Benjamins.

Hogg, Richard M. 1992. *A Grammar of Old English. Volume 1: Phonology*. Blackwell.

Luick, Karl. 1914–1940. *Historische Grammatik Der Englischen Sprache*. Tauchnitz.

Ringe, Don, and Ann Taylor. 2014. *The Development of Old English*. Vol. 2. A Linguistic History of English. Oxford University Press.